

Department of Electronics Engineering
B.Tech. III Sem. V

End-Semester Examination, November – 2024

Digital Signal Processing (EC 301)

U22EC004

Time: 9:30AM

Duration: 3 Hours.

Total Marks: 50

Instructions:

1. Attempt all questions.
2. Assume necessary data if required.
3. Total number of questions are 10.

Q-1	Explain the up-sampling operation with suitable example. How to remove spectral images introduced by up-sampling?	(02+03) (CO2)
Q-2	Calculate the impulse response $h(n)$ of a Low Pass FIR Filter for the given specifications using either Fourier series method or Fourier transform/Sampling method: (a) cut-off frequency: 500 Hz (b) sampling frequency: 2000Hz (c) order of the filter: 10 (d) length of the filter: 11	(05) (CO4)
Q-3	Analog filter transfer function $H(s)$ is given by following equation. Calculate the digital filter transfer function using impulse invariant design method (consider sampling period=1) $H(s) = \frac{1}{(s+5)(s+8)}$	(05) (CO3)
Q-4	Compute the eight point DFT of the given sequence $x(n)$ using decimation in time (DIT) algorithm. Draw the butterfly structure for the algorithm and mention output at each stage. Mention the required ordering of the given sequence at the input of the algorithm and also the order of the resulted DFT sequence. If required, rearrange the resulted DFT sequence to obtain the output sequence in natural order. $x(n)$ is given by $x(n) = (1, -1, 1, 0, 2, -1, 2, 1)$	(06) (CO4)
Q-5	(i) Explain the different IIR filter structures such as Direct Form I, Direct Form II, and Transposed structures with suitable example. (ii) Draw the circuit diagram of IIR filter for Parallel and Cascade structures for the given transfer function: $H(z) = \frac{1+\frac{1}{3}z^{-1}}{1+\frac{7}{10}z^{-1}+\frac{1}{10}z^{-2}}$	(03+03) (CO1)

Q-6	Design a Butterworth analog low pass filter for the following specifications: (i) pass-band gain required: 0.9 (ii) frequency upto which pass-band gain must remain steady: 100 rad/s (iii) amount of attenuation required: 0.4 (iv) frequency from which attenuation must start: 200 rad/s.	(05) (CO4)
Q-7	A signal $x(n)$ is given by following equation. $x(n) = (1/2)^{ n-1 } + (1/2)^{ n+1 }$ Calculate the DTFT of the signal $x(n)$. Also calculate the magnitude and phase response.	(05) (CO3)
Q-8	What are significance of window function in FIR filter design. Draw the time domain and frequency domain signal of rectangular, Hamming and Hanning Window function. Explain the advantage and disadvantage of above window functions.	(05) (CO1)
Q-9	The low pass filter with cut-off frequency of 0.2π is given by following system function $H(z) = \frac{0.5(1 + z^{-1})}{1 - 0.8z^{-1}}$ Use frequency transformation technique to transform the above low pass filter into a band pass filter with upper and lower cut-off frequencies 0.6π and 0.4π respectively.	(05) (CO3)
Q-10	Consider a discrete time system with input-output relationship is given by $y(n) - \frac{1}{2}y(n-1) = x(n)$ The input $x(n)$ is given by $x(n) = (\frac{1}{3})^n u(n)$. Calculate the output $y(n)$.	(03) (CO2)